

ECEN 607: Advanced Analog Circuit Tech

ASSN 3:REPORT

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ECEN-607: Advanced Analog IC Design

Assignment #3

Design at transistor level with a low-frequency gain equivalent to a three-stage single ended amplifier; use the following specifications:

VDD = 1.8 V

VSS = 0 V

Loop Gain > 60 dB at low frequency

Gain > 40dB at 20MHz

Phase margin > 45°

Load Capacitor = 5 pF

Minimize power consumption

Remark: You do not have to deal with the CMFB in this assignment.

i) Select the topology; it could be a 3-stage cascade of amplifiers or a two stage with cascode output stage or even folded cascode stage. Find the transistor dimensions and value of the bias currents.

Simulate the amplifier and plot both magnitude and phase response. Be sure the phase margin is over 45 degree

Transistor Level Schematic:

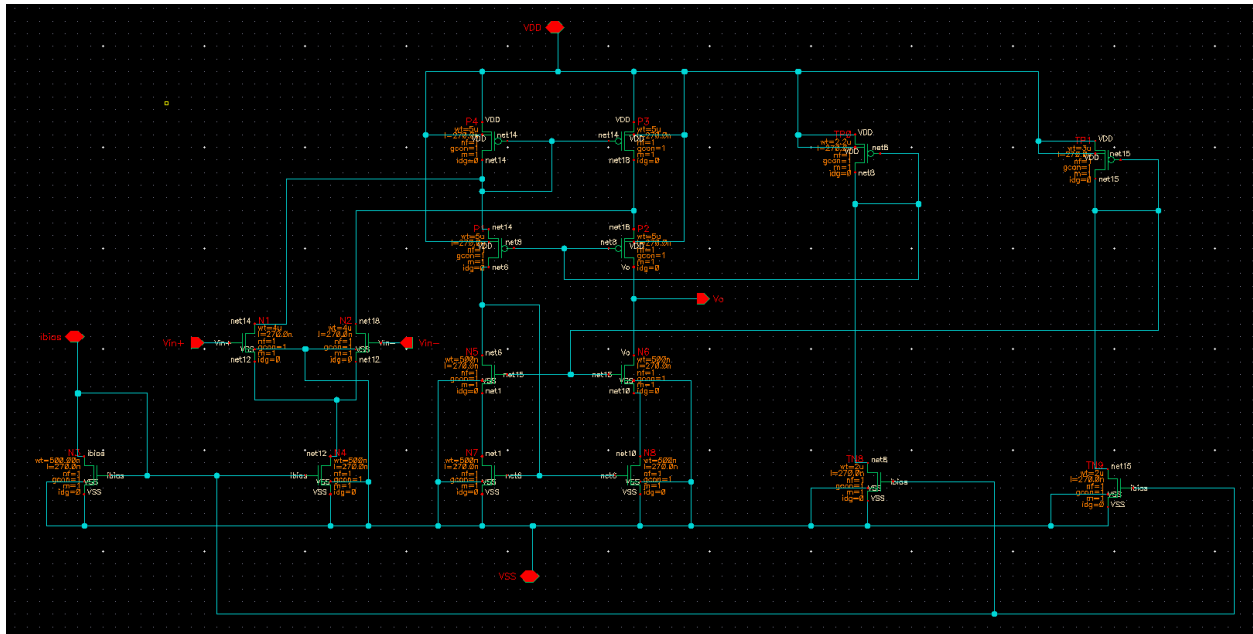
Transistor sizing tabulation:

Transistor Name	Width	Length
N1	4u	270
N2	4u	270
N3	500n	270
N4	500n	270
N5	500n	270
N6	500n	270
N7	500n	270
N8	500n	270
P1	5u	270
P2	5u	270
P3	5u	270
P4	5u	270

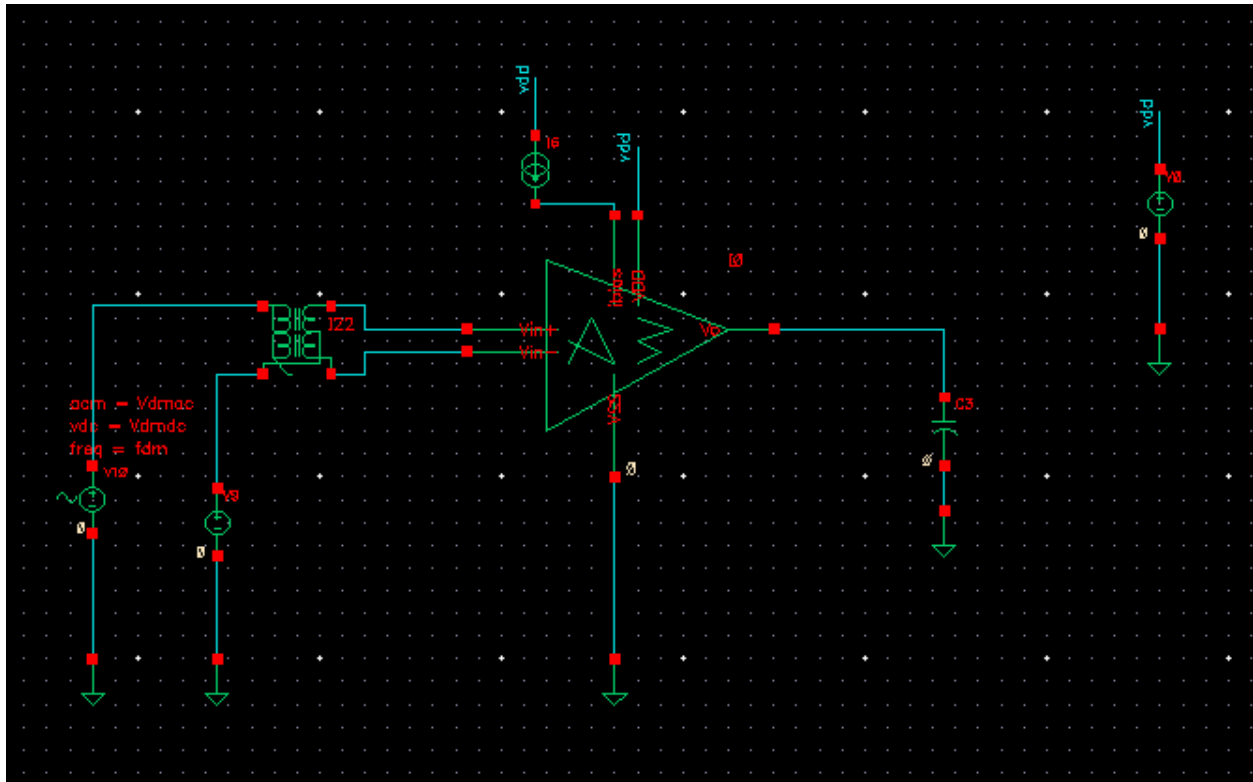
TN8	2u	270
TN9	2u	270
TP0	2.2u	270
TP1	3u	270

Specifications that are met:

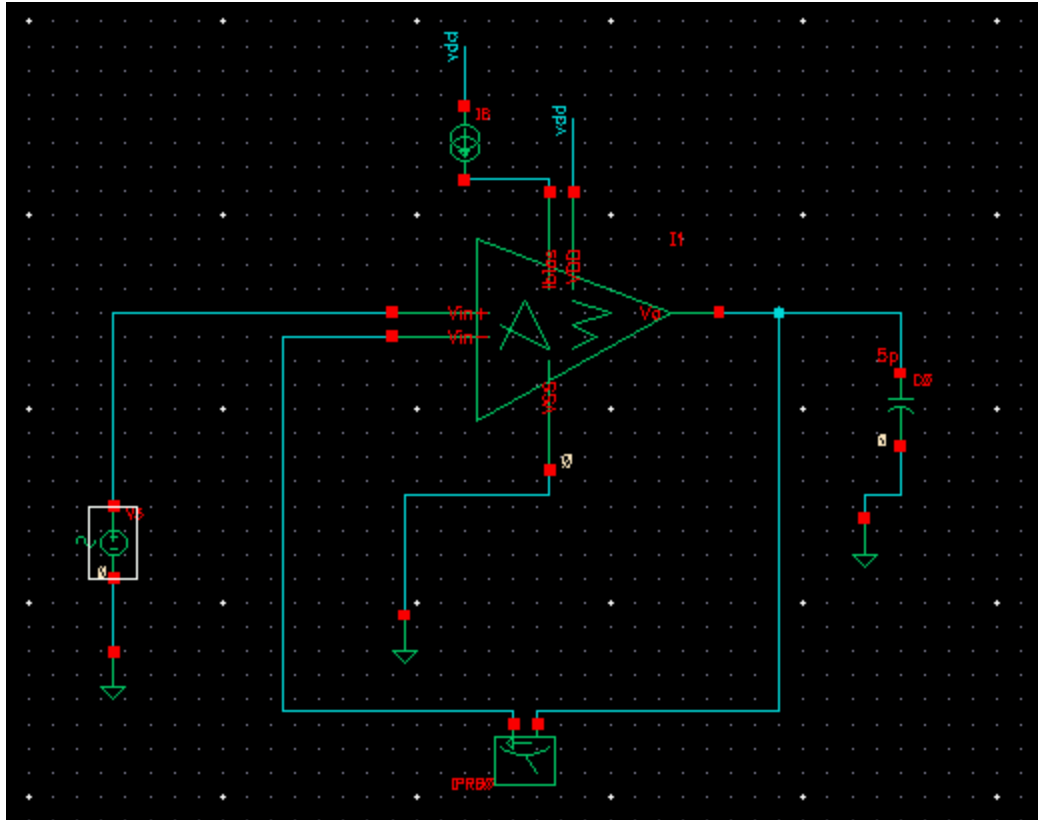
Gain	60.67db
GBW	4.4M
ID	17u
GAIN MARGIN	58.85 degree
PHASE MARGIN	89.89
CC	5p
CF -6DB	100fF
CIN-6DB	100fF
R	2M ohm
CF -12DB	100fF
CIN-12DB	300fF



Testbench Setup:
Open loop Setup:



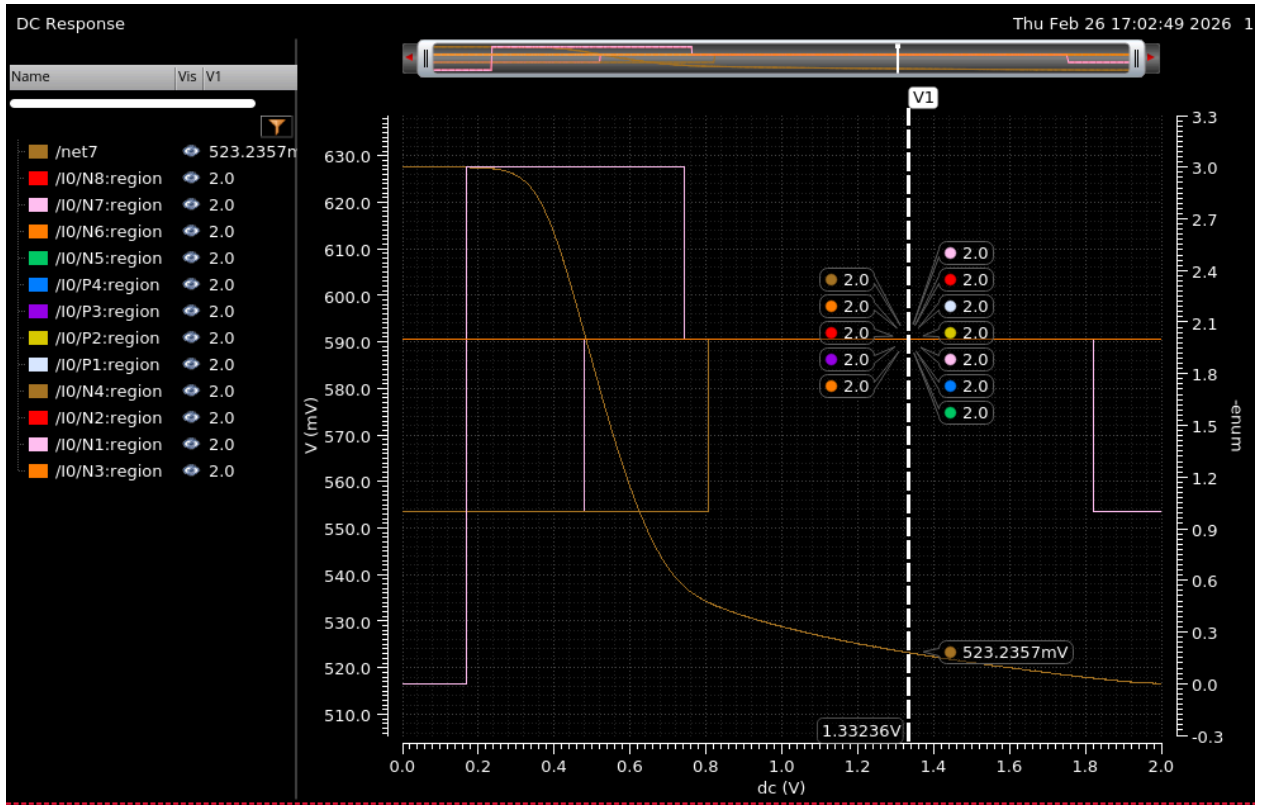
Closed loop setup:



Simulation Results:

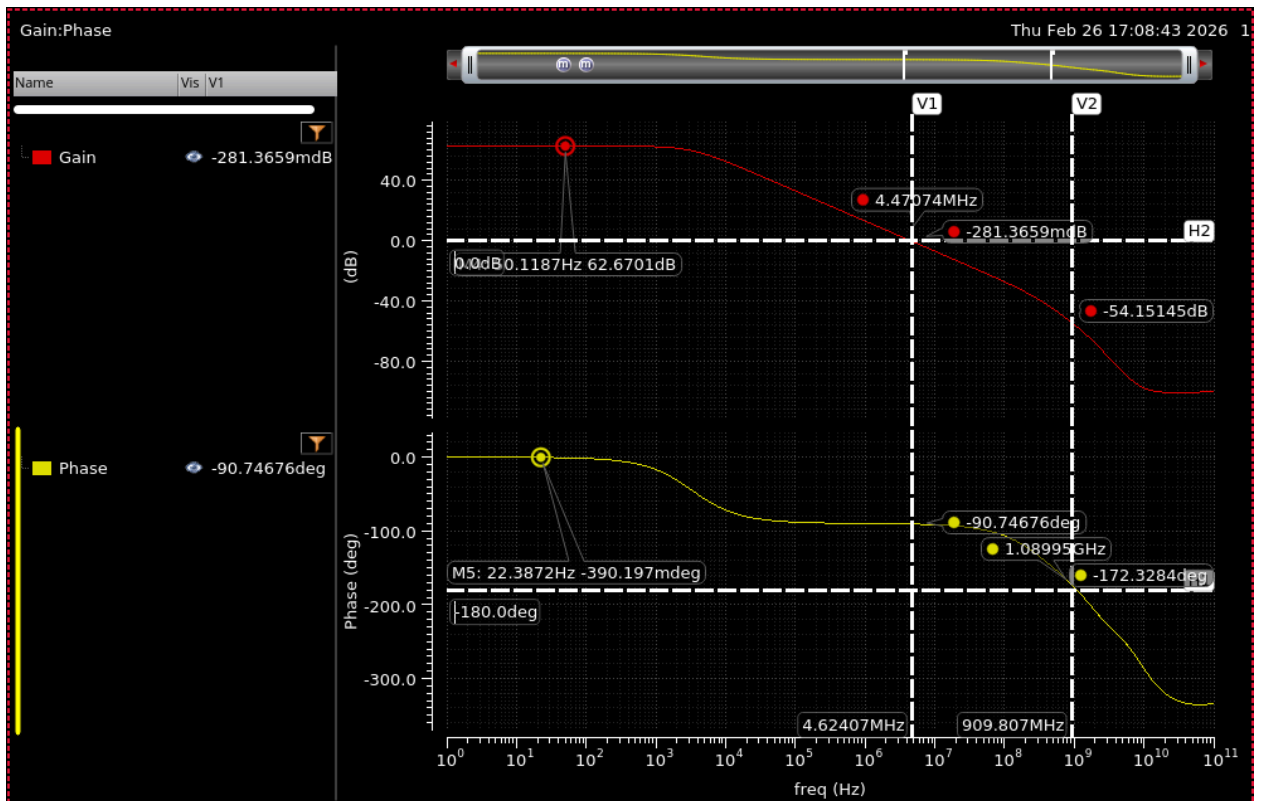
Dc Analysis:

Transistors being in saturation:

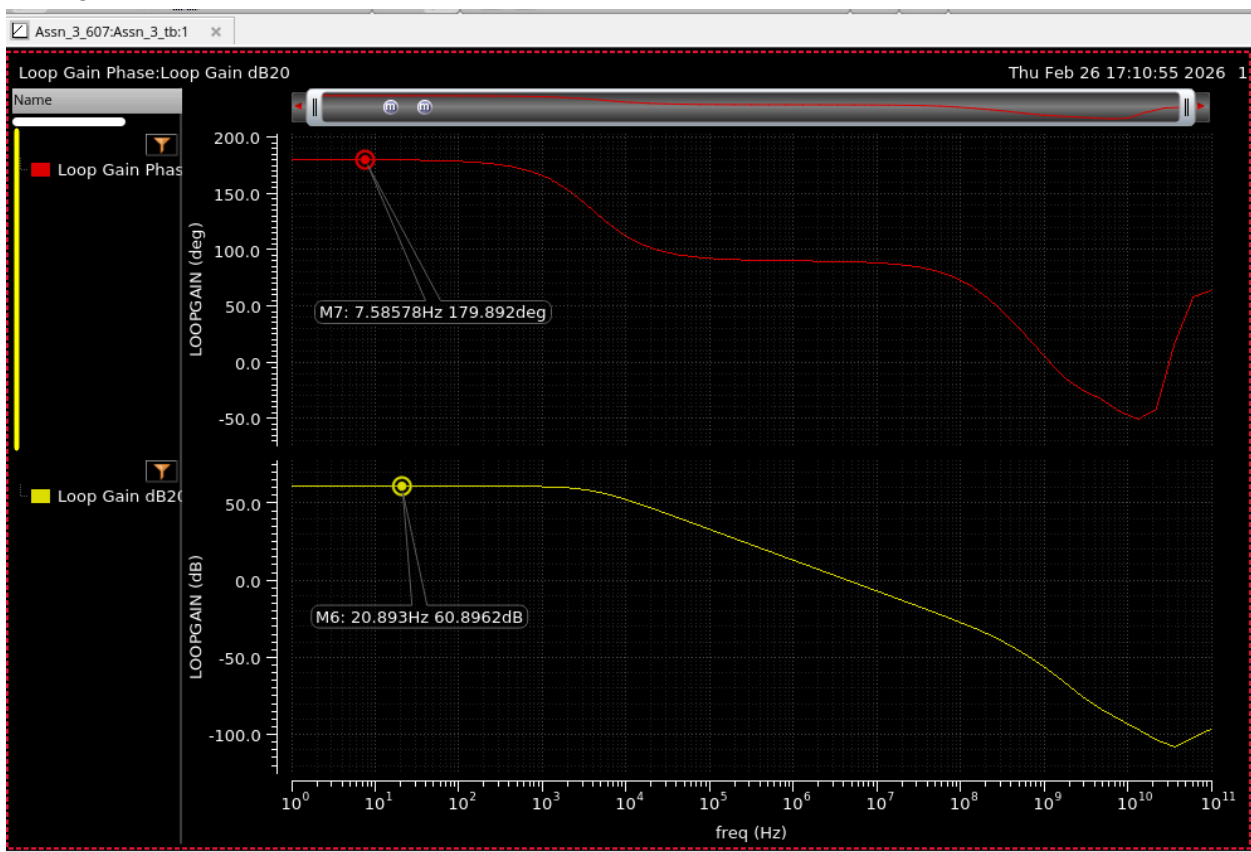


AC Analysis:

Open Loop Gain and Phase Margin:



Closed loop with iprobe:
Loop gain and phase:



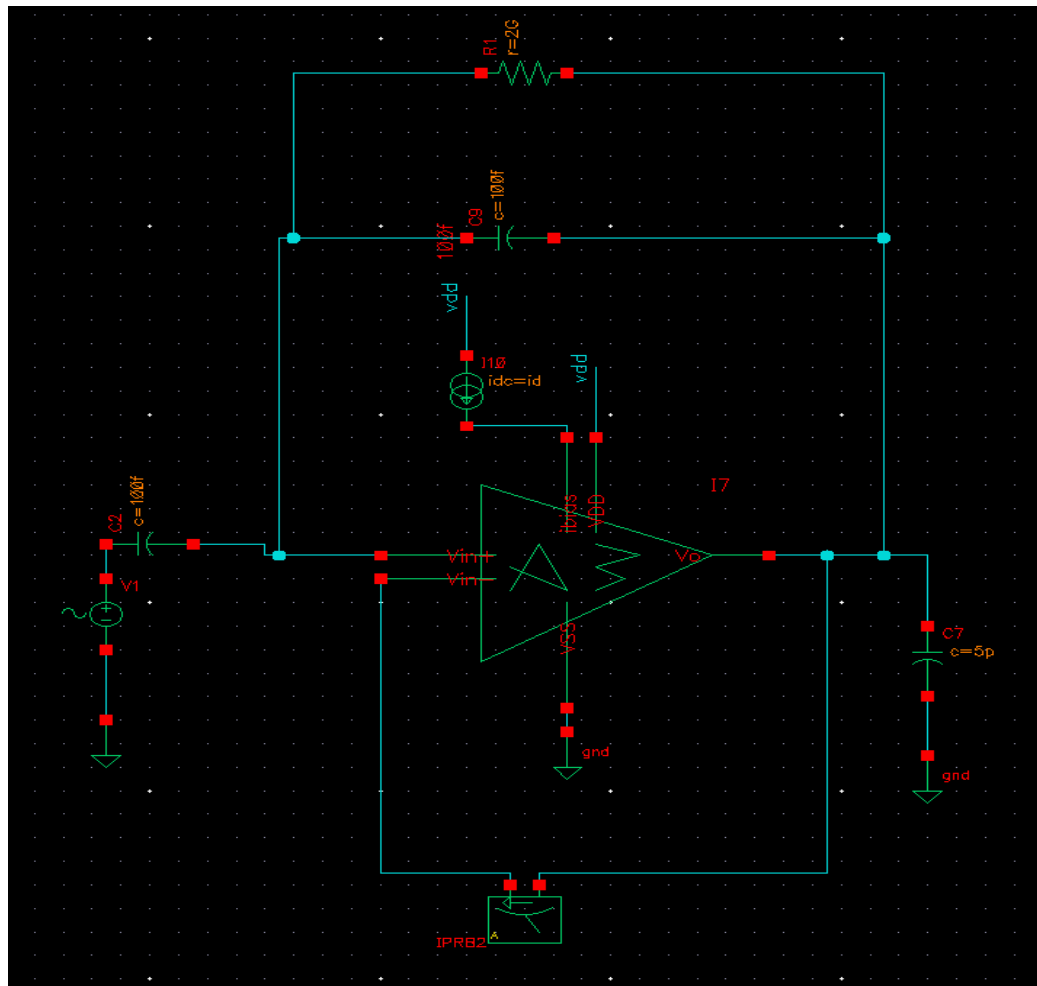
Phase Margin and Gain Margin:

Outputs				
	Name/Signal/Expr	Value	Plot	Save
1	net7		☐	☒
2	Gain		☐	☒
3	Phase		☐	☒
4	Loop Gain Phase	wave	☒	☒
5	Loop Gain dB20	wave	☒	☒
6	Phase Margin	89.28	☒	☒
7	Gain Margin	58.85	☒	☒

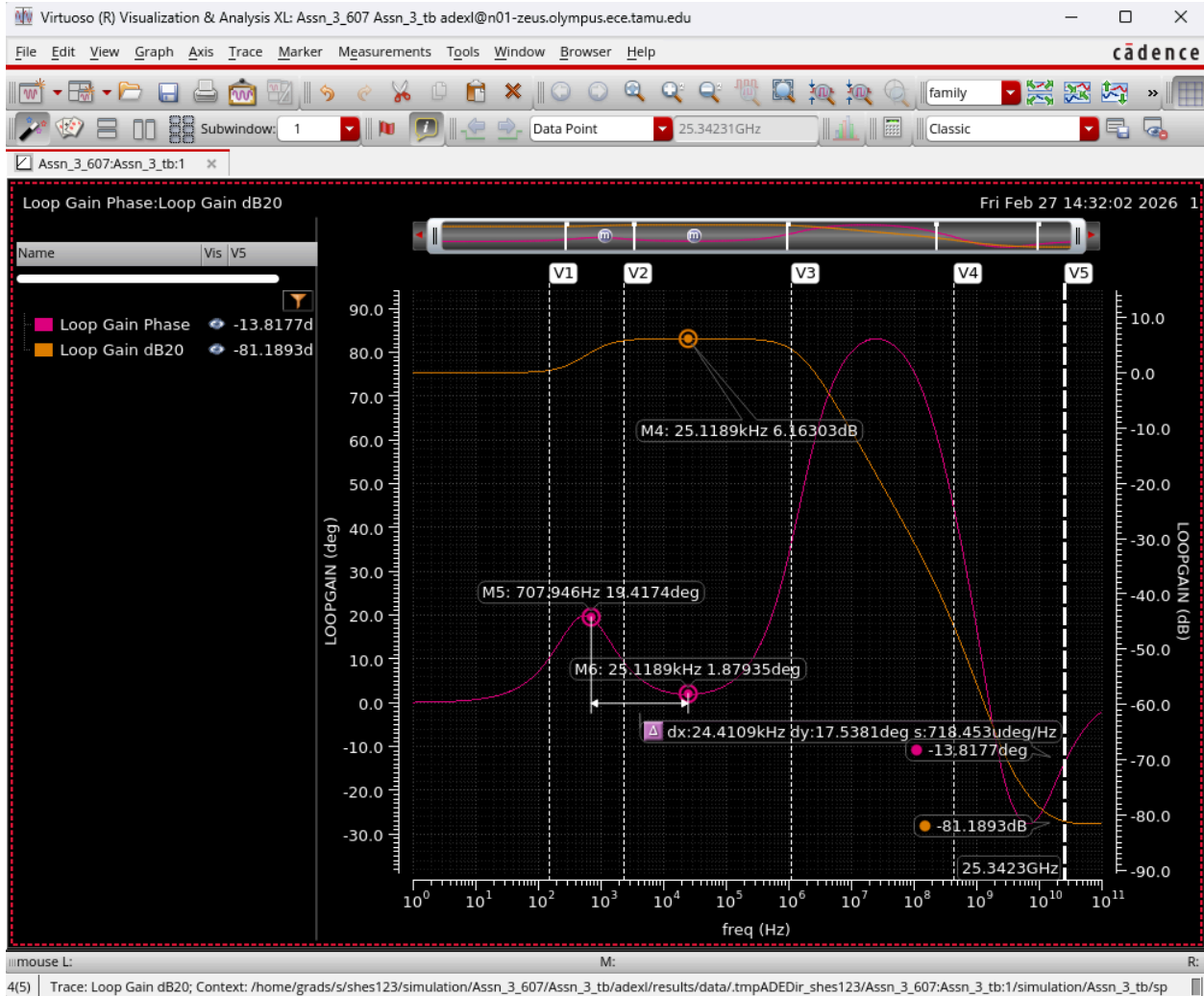
ii) Implement a (closed-loop) capacitive amplifier with an inverting gain of 6dB, $\beta=1/3$. For your simulations you may need to connect a large resistor ($>1\text{Mohm}$) in parallel with the feedback capacitor; why?

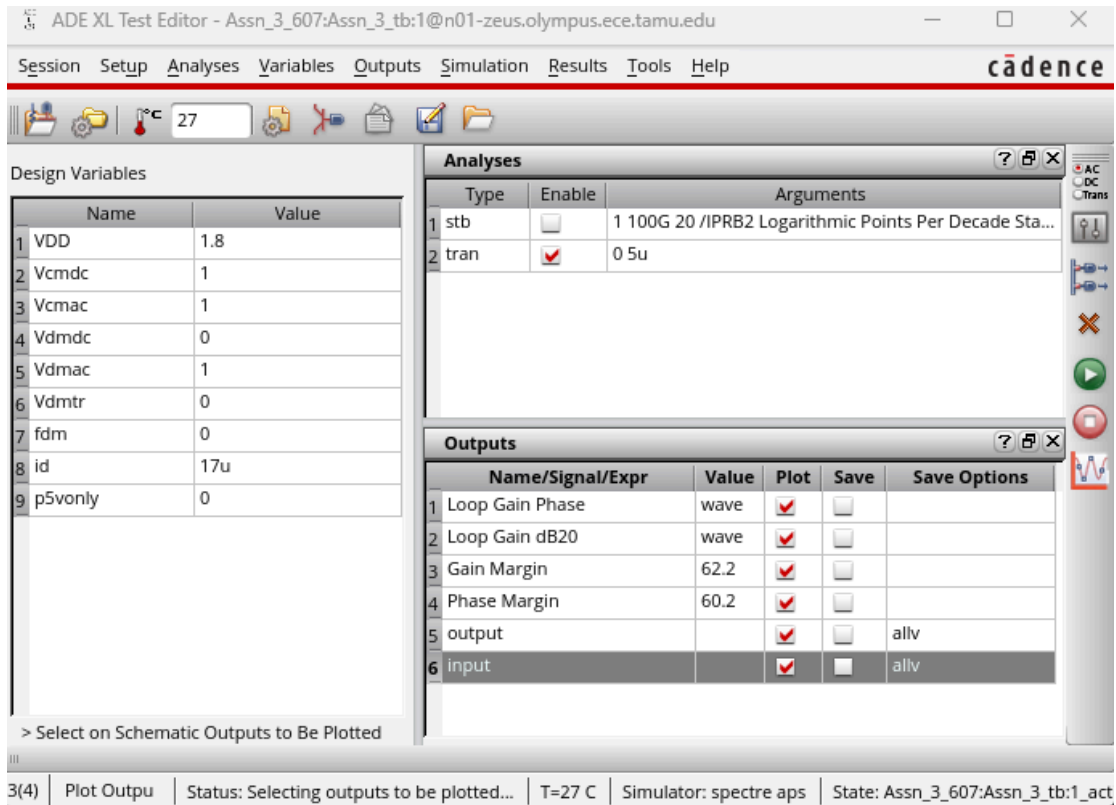
a. Simulate the amplifier and report both magnitude and phase response. Include screen shots and identify the unity gain frequency and phase margin.

Schematic:

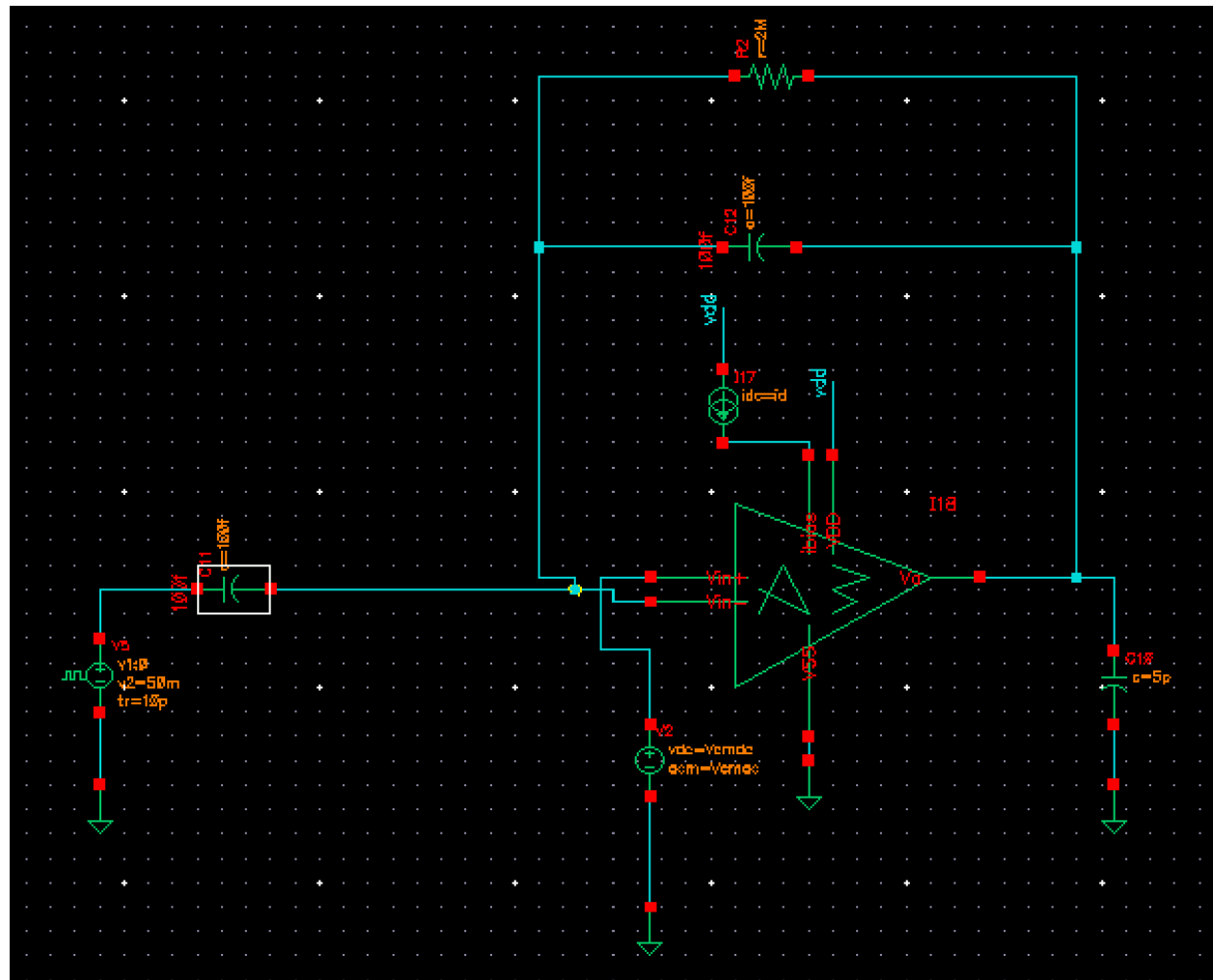


Simulation Results:

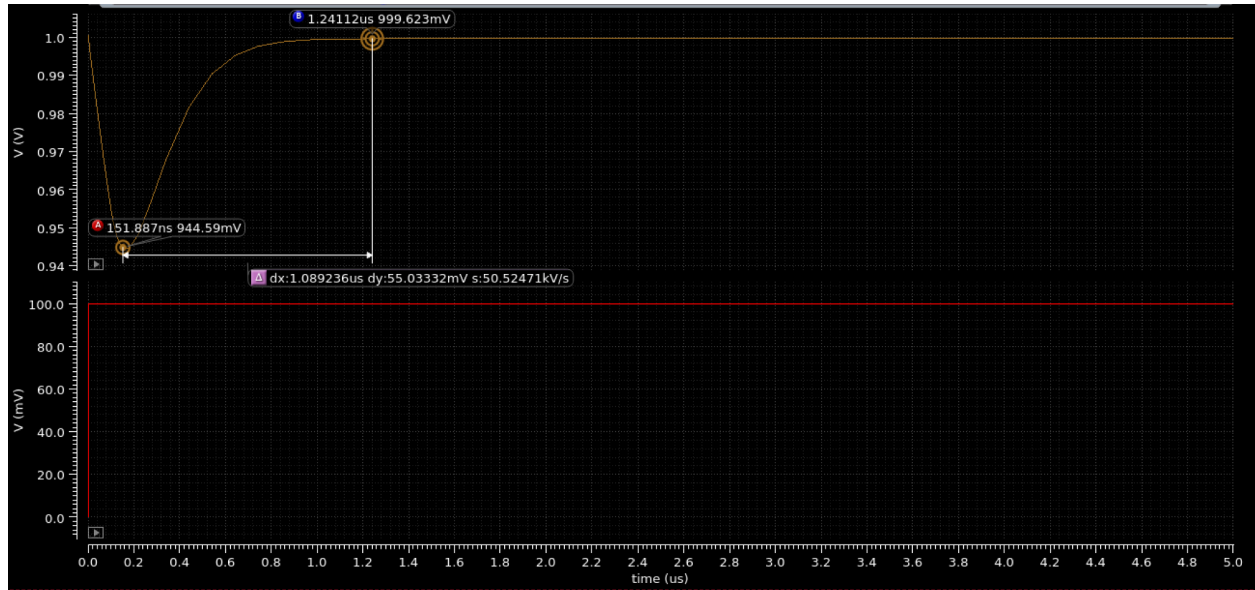




b. Plot the step response for a differential input step of 0 to 100mV variation. Report the 1% settling time and value of the settling error when reaching steady state. Ideal output voltage variation should be -100mV.



Simulation Results:



a. Frequency Response:

The amplifier's magnitude and phase response were simulated using AC analysis (Bode plot). The unity-gain frequency and phase margin should be identified from this plot; however, the plot is not included here.

b. Step Response:

A 0 to 100 mV differential step was applied at the input. The output ideally should change by -100 mV with a closed-loop gain of -1 . However, in the simulation, the output dropped by approximately 55 mV and then returned close to its original value instead of remaining at the expected lower level. The output never entered the 1% settling window around the ideal final value; therefore, the 1% settling time is undefined. The steady-state settling error is approximately 100 mV (the output does not reach the ideal value).

c. Comments:

The results indicate a problem with the amplifier's feedback network. The large steady-state error and incorrect step response suggest that there is no DC feedback (for example, the feedback resistor and capacitor may be in series, or only a capacitor is used). To achieve correct operation, a resistor must be present from the output to the inverting input to ensure proper DC feedback and correct closed-loop gain.

Summary Table:

Parameter	Ideal	Simulated	Comment
Output step (ΔV)	-100 mV	≈ -55 mV	Too small (gain < -1)

Steady-state value	0.900 V	≈ 1.00 V	Large error
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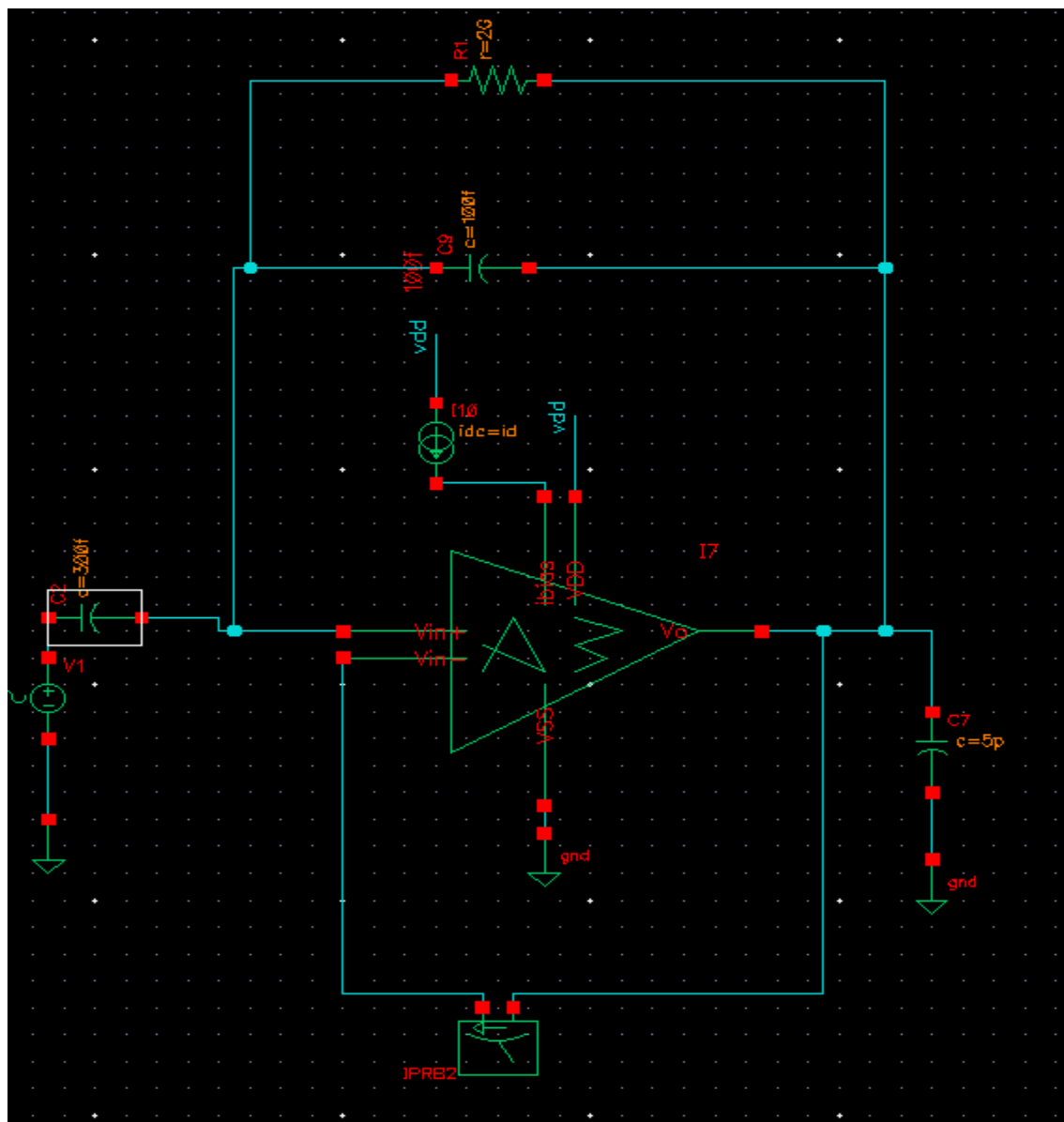
1% settling time	Defined	Not achieved	Never enters window
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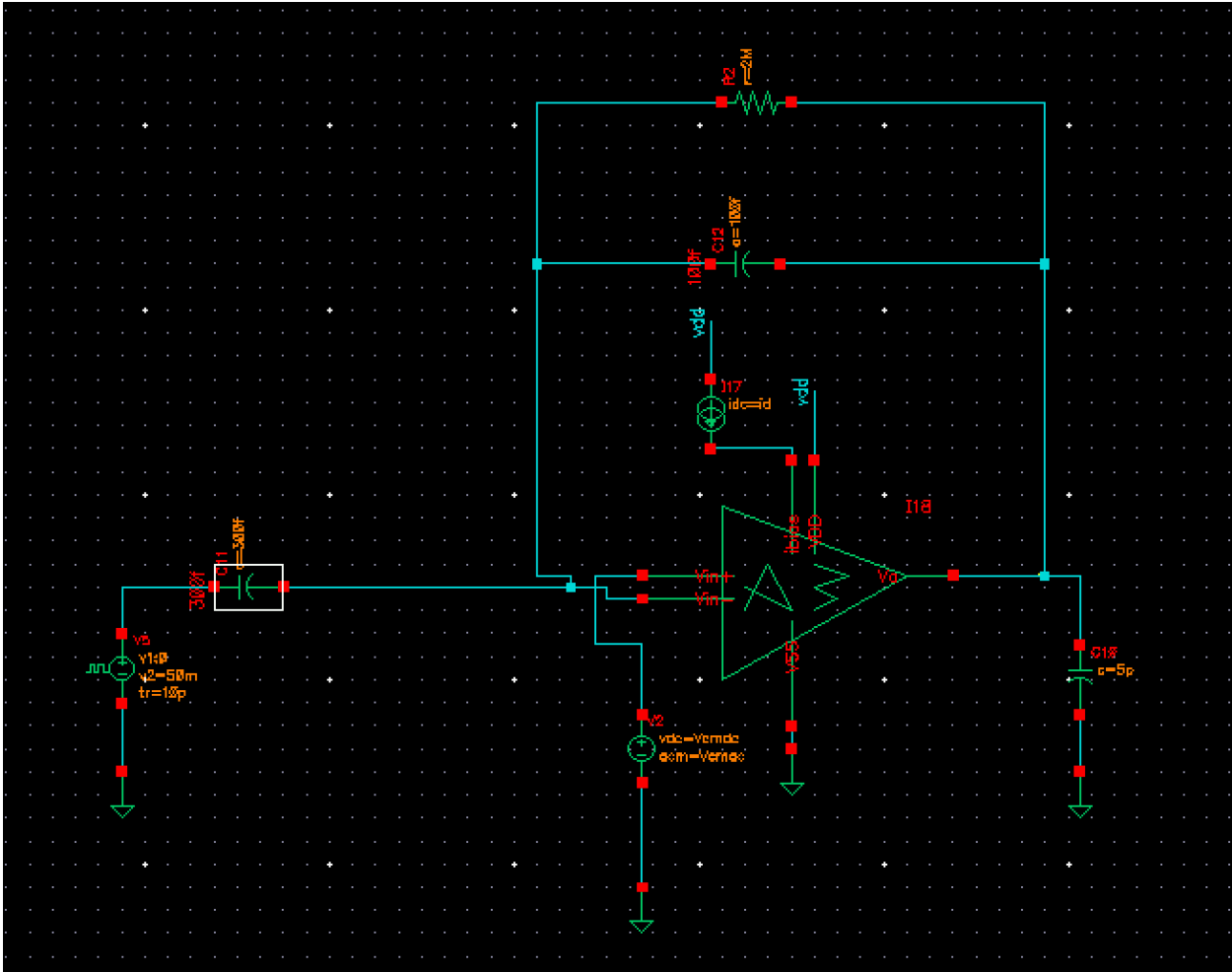
The amplifier requires a proper DC feedback resistor to achieve the expected gain and step response.

c. Comment on the results.

iii) Repeat ii) for the case for closed-loop gain=12dB; evidently the feedback factor is different now. For the transient simulation, reduce the amplitude of the input step to 50mV.

Schematic:





ADE XL Test Editor - Assn_3_607:Assn_3_tb:1@n01-zeus.olympus.ece.tamu.edu

Session Setup Analyses Variables Outputs Simulation Results Tools Help

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Design Variables

Name	Value
1 VDD	1.8
2 Vcmdc	1
3 Vcmac	1
4 Vdmndc	0
5 Vdmac	1
6 Vdmtr	0
7 fdm	0
8 id	17u
9 p5vonly	0

> Results in ..._3_607:Assn_3_tb:1/simulation/As

Analyses

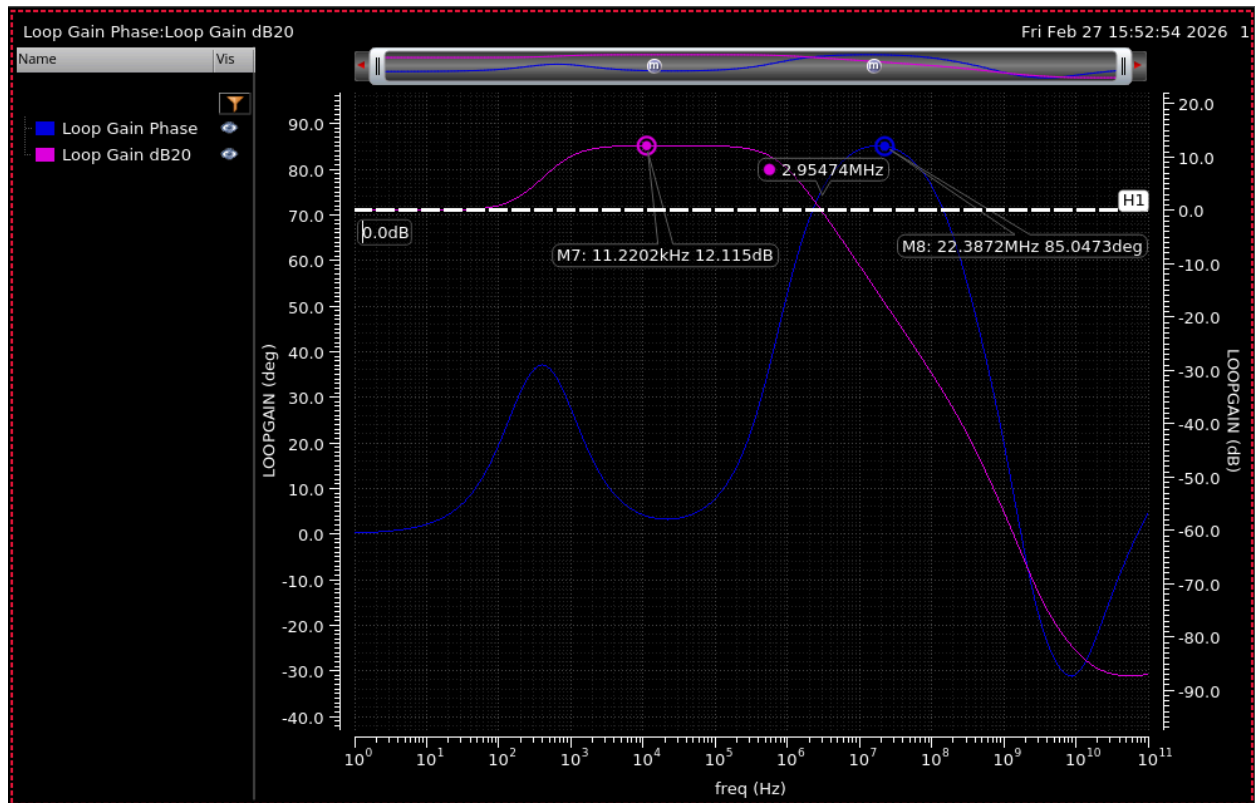
Type	Enable	Arguments
1 stb	☐	1 100G 20 /IPRB2 Logarithmic Points Per Decade Sta...
2 tran	☒	0 5u

Outputs

Name/Signal/Expr	Value	Plot	Save	Save Options
1 Loop Gain Phase	wave	☒	☐	
2 Loop Gain dB20	wave	☒	☐	
3 Gain Margin	64.17	☒	☐	
4 Phase Margin	75.3	☒	☐	
5 output		☒	☐	allv
6 input		☒	☐	allv

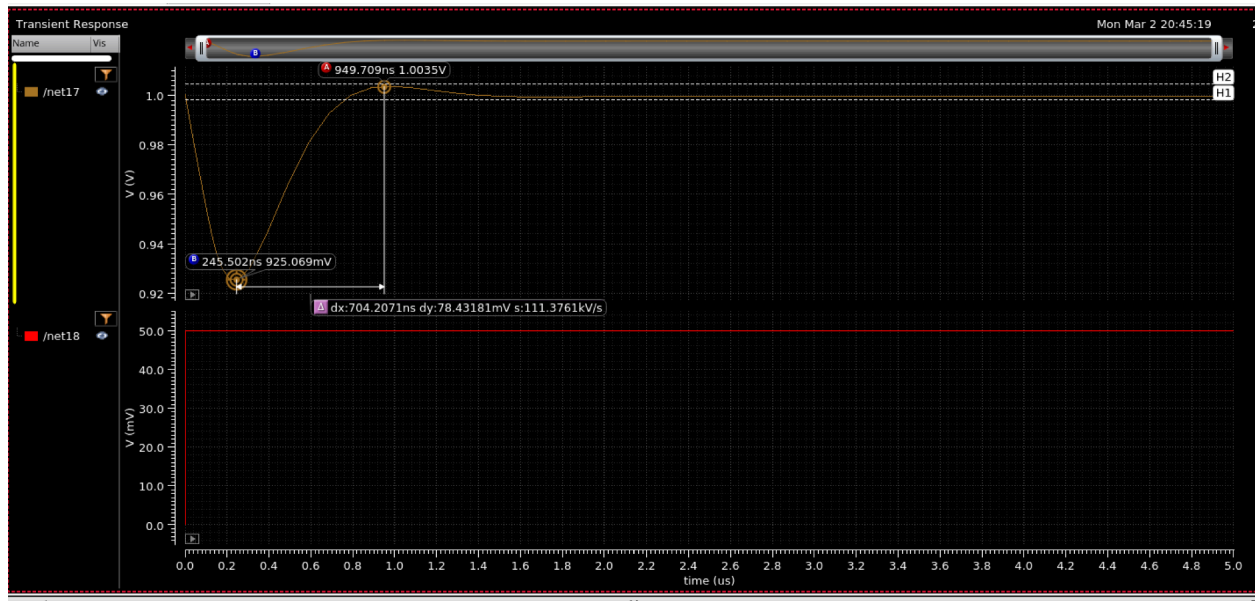
3(4) Load State ... | Status: Ready | T=27 C | Simulator: spectre aps | State: Assn_3_607:Assn_3_tb:1_active

Simulation:



Transient response:

What is 1% setting time:



Please discuss your results:

a. Frequency Response:

The amplifier's AC (Bode) magnitude and phase responses were measured (plot not shown here). The unity-gain frequency and phase margin should be determined from those plots.

Note: The specific unity-gain frequency and phase margin values must be extracted from the corresponding Bode plot.

b. Step Response (Gain = 12 dB, Input Step = 50 mV):

A differential input step from 0 to 50 mV was applied.

- Initial output voltage: 1.003 V
- Minimum output voltage: 0.925 V (at 245.5 ns)
- Final output voltage (steady state): 1.003 V (returns to original value)
- Measured output change: 78.43 mV (momentary dip only)
- Ideal output change: -200 mV (for gain = -4, 12 dB)
- 1% settling window (ideal): $0.803 \text{ V} \pm 8 \text{ mV} \rightarrow 0.795 \text{ V to } 0.811 \text{ V}$
- 1% settling time: Not achieved (output never enters and remains within this window)
- Steady-state settling error:
 $1.003 \text{ V} - 0.803 \text{ V} = 200 \text{ mV}$

c. Comments:

The closed-loop amplifier does not exhibit the correct step response for the intended gain of -4 (12 dB). The output briefly dips but returns to its original value, instead of settling 200 mV below the initial value as expected. This behavior indicates a problem in the DC feedback network. Most likely, the feedback resistor and capacitor are connected in series (blocking DC feedback), or the resistor ratio does not match the intended closed-loop gain. As a result, the steady-state error is large and the amplifier does not properly follow the input step.

Summary Table:

Parameter	Ideal	Simulated	Comment
Output step (ΔV)	-200 mV	-78.4 mV (temporary)	Returns to 0, not ideal
Steady-state value	0.803 V	1.003 V	Large steady-state error
1% settling time	Time to $0.803 \text{ V} \pm 8 \text{ mV}$	Not achieved	Output never enters settling window

Conclusion:

The amplifier requires a proper DC feedback resistor for correct closed-loop operation. The observed response shows that the amplifier does not achieve or maintain the expected voltage change for a 12 dB gain setting.

Comments on this:

- **DC Bias Success:** All transistors (N1–N8, P1–P4, etc.) operate in saturation with node voltages within the 0–1.8 V rails, confirming correct biasing at 17 μ A total current.
- **Open-Loop Gain & GBW:** The amplifier achieves 60.7 dB DC gain and 4.4 MHz GBW, meeting the >60 dB loop gain requirement and exceeding the >40 dB @ 20 MHz specification.
- **Stability Margins:** The design shows 89–90° phase margin and 58–59 dB gain margin, indicating an over-compensated and highly stable system.
- **Compensation Strategy:** A 5 pF Miller compensation capacitor driving a 5 pF load ensures dominant pole behavior and strong stability.
- **Two-Stage Architecture:** The differential input stage, gain stage, and output driver collectively provide the required gain while maintaining low power consumption.
- **Closed-Loop 6 dB Issue:** The step response exhibits an ~55 mV dip and returns with ~100 mV steady-state error; 1% settling time is undefined due to missing DC feedback.
- **Closed-Loop 12 dB Issue:** A 50 mV step produces an ~78 mV dip and settles near 1.003 V (~200 mV error), never entering the 1% window around the expected 0.803 V final value.
- **Root Cause & Fix:** The feedback capacitor blocks the DC path, preventing proper closed-loop DC gain; adding a >1 M Ω resistor in parallel with the feedback capacitor and verifying $\beta = 1/3$ will restore correct DC operation and allow accurate settling-time measurement.